

LIVER

Regular anatomy notes - source diagrams, clinical points, and exam-focused layout

What the liver is

- The liver is the largest gland of the body and occupies most of the right upper abdomen, lying just below the diaphragm.
- It performs metabolic, synthetic, detoxifying, storage, hematopoietic (fetal), and bile-secreting functions.
- It is a highly vascular organ receiving venous nutrient-rich blood from the portal vein and arterial blood from the hepatic artery.

Why it matters clinically

- Its superior position under the ribs explains why liver enlargement is palpated below the right costal margin.
- Its segmental architecture permits partial hepatectomy and segmental resection.
- Portal venous drainage and hepatic venous outflow are central to understanding portal hypertension and liver surgery.

life.

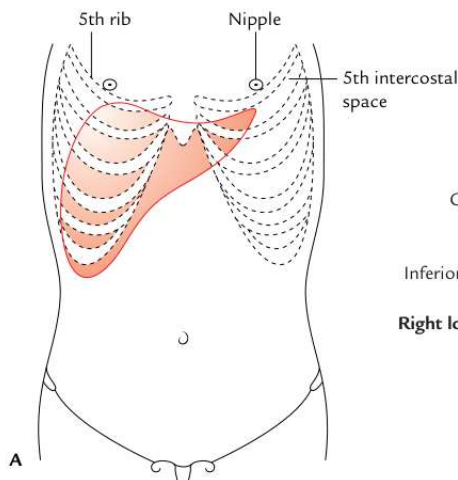


Fig. 8.1).

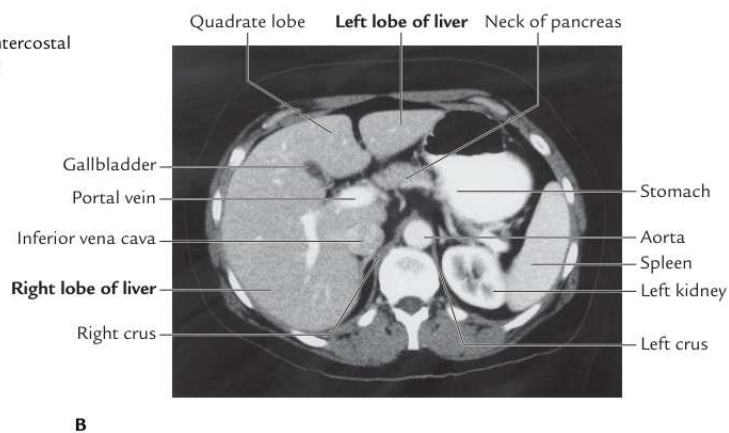


Fig. 8.1 Location of the liver: A, surface projection of the liver as seen from the front; B, CT scan of the abdomen showing the location of the liver. (Source: Fig. 4.85B, Page 286, *Gray's Anatomy for Students*, Richard L Drake, Wayne Vogl, Adam WM Mitchell. Copyright Elsevier Inc. 2005, All rights reserved.)

Source diagram: surface projection and CT location of the liver.

The liver lies mainly beneath the right costal margin and diaphragm. On surface projection, it extends from the right 5th rib region across the epigastrium toward the left 5th intercostal space. The CT image reinforces its relation to the gallbladder, portal vein, IVC, stomach, spleen, kidney, and diaphragm.

1. Location, Shape, Size, and Colour

Location

The liver lies in the right hypochondrium, epigastrium, and partly in the left hypochondrium. It is mostly covered by ribs and costal cartilages and lies immediately below the diaphragm. The superior surface fits into the dome of the diaphragm, while the visceral surface faces downward, backward, and left.

Shape and colour

- Wedge-shaped; resembles a four-sided pyramid with its base to the right and apex to the left.

Size and weight

- Adult male: roughly 1.4-1.8 kg; adult female: roughly 1.2-1.4 kg.

- Has two major surfaces: diaphragmatic and visceral.
- Colour is red-brown because of its rich blood supply.

- Newborn: about 1/18th of body weight; adult: about 1/40th of body weight.
- Liver is proportionately large in newborns because of its fetal hematopoietic function.

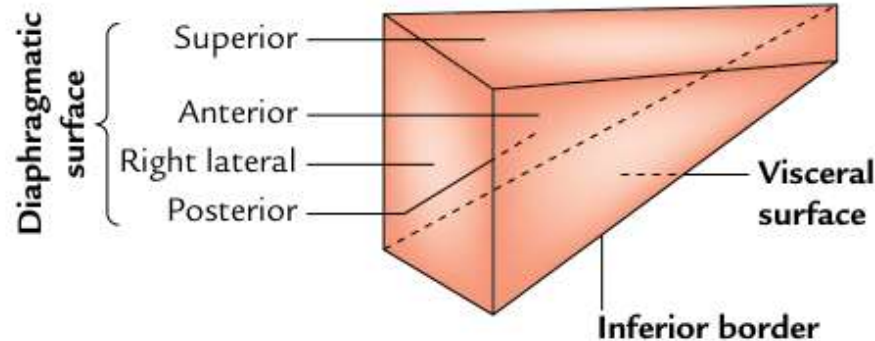


Fig. 8.2 Schematic diagram to show shape (wedge shape) and surfaces of the liver.

Source diagram: wedge shape and surfaces of the liver.

This schematic is useful for remembering that the diaphragmatic surface is extensive and convex, while the visceral surface is flatter and lies inferiorly. The sharp inferior border marks the meeting line of the two surfaces anteriorly.

How to orient this diagram

- The base of the wedge is on the right side and the apex points to the left.
- The visceral surface is the inferior surface; it carries impressions of abdominal viscera.
- The diaphragmatic surface is continuous over superior, anterior, right lateral, and posterior aspects.

Quick memory cue

- Dome = diaphragm, flat = viscera.
- Inferior border is sharp medially and helps in surface marking and palpation.
- A liver edge felt far below the costal margin may indicate enlargement or descent with inspiration.

Clinical orientation note

During abdominal examination, the examiner asks the patient to inspire deeply. The liver descends with the diaphragm, so the inferior border becomes easier to feel below the right costal margin. A firm, irregular, or tender edge is clinically significant and should always be correlated with the surface projection shown above.

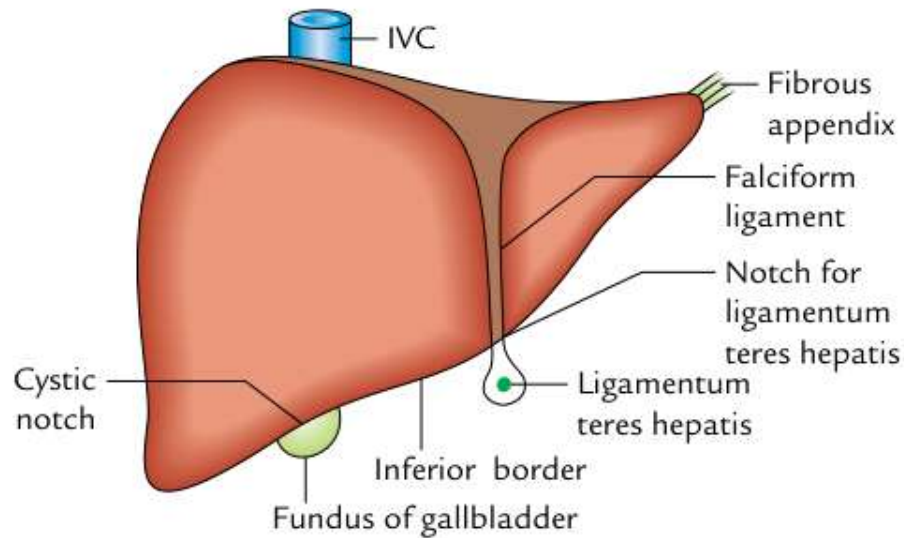


Fig. 8.3 Liver—viewed from the front.

Source diagram: liver viewed from the front.

From the front, the right lobe is much larger than the left. The falciform ligament is attached anteriorly, the fundus of the gallbladder projects below the inferior border, and the IVC lies posteriorly.

Front-view labels to remember

The cystic notch marks the position of the gallbladder at the inferior border. The notch for ligamentum teres lies nearer the median plane. The falciform ligament reaches the anterior surface but does not represent the true functional plane of division of the liver.

2. External Features and Surfaces

Diaphragmatic surface

This surface is convex and extensive. It faces upward, forward, to the right, and backward. It is related to the diaphragm and, through the diaphragm, to the pleura, lungs, pericardium, and heart. Its posterior part includes the bare area, which is not covered by peritoneum.

Visceral surface

The visceral surface is relatively flat or concave and faces downward, backward, and to the left. Important features include the fossa for the gallbladder, fissure for ligamentum teres hepatis, fissure for ligamentum venosum, groove for the IVC, and porta hepatis.

Border and notches

- The inferior border is the sharp anterior border where diaphragmatic and visceral surfaces meet.
- The cystic notch corresponds to the fundus of gallbladder.
- The notch for ligamentum teres lies near the median plane.

Why surface features matter

- Surface grooves and fissures guide identification of lobes during practical examination.
- They also mark sites for peritoneal reflections and the pathway of fetal remnants.
- Porta hepatis is the key landmark for vessels, ducts, lymphatics, and nerves.

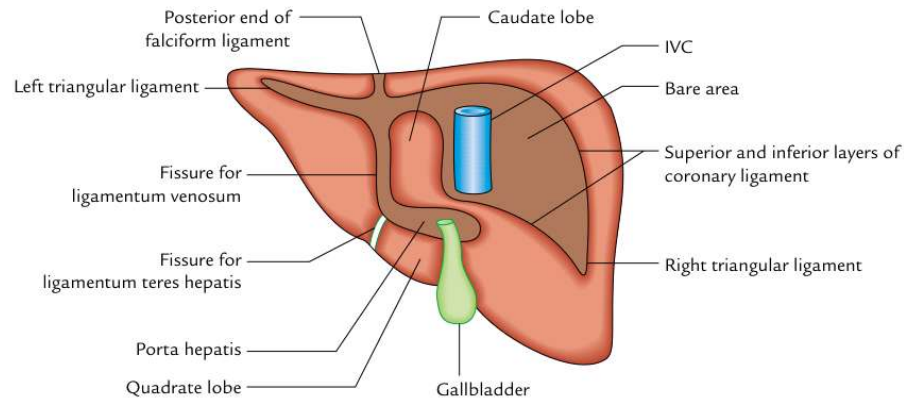


Fig. 8.4 Liver—viewed from behind.

Source diagram: liver viewed from behind.

The posterior view shows the bare area, groove for the IVC, fissures for ligamentum venosum and ligamentum teres, caudate and quadrate lobes, porta hepatis, and peritoneal reflections. It is especially important for peritoneal relations.

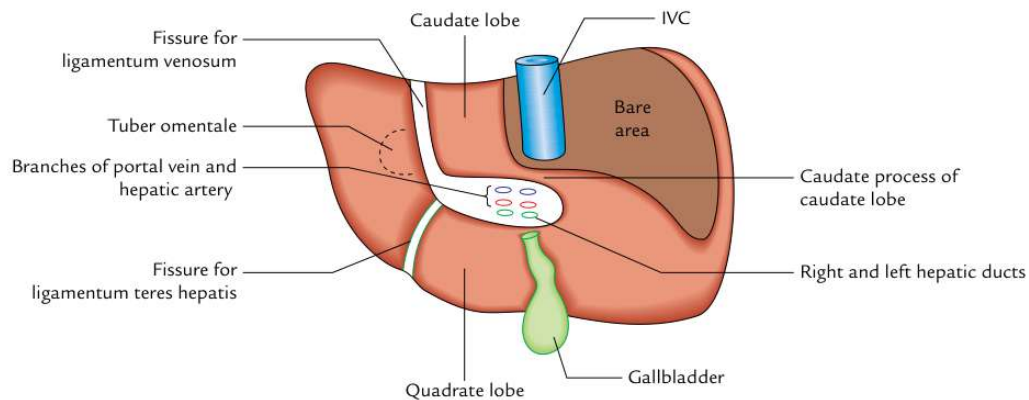


Fig. 8.5 Liver—viewed from below.

Source diagram: liver viewed from below.

The inferior view shows the visceral surface. The porta hepatis lies between the caudate and quadrate lobes and transmits the portal triad structures. The gallbladder lies in its fossa, while the IVC is lodged in the posterior groove.

Notable Features of the Surfaces

Surface	Important features
Diaphragmatic surface	Bare area of liver; groove for IVC; fissure for ligamentum venosum; attachment of coronary, triangular, and falciform ligaments.
Visceral surface	Fissure for ligamentum teres; fossa for gallbladder; porta hepatis; fissure for ligamentum venosum; impressions of stomach, duodenum, colon, kidney, and suprarenal gland.
Inferior border	Sharp medially, rounded laterally; presents the cystic

notch and notch for ligamentum teres; surface landmark crosses near the transpyloric plane.

3. Lobes of the Liver

Anatomical lobes

- Based mainly on external surface features.
- On diaphragmatic surface: right and left lobes separated by the falciform ligament.
- On visceral surface: right, left, caudate, and quadrate lobes are identified by fissures, fossae, porta hepatis, and groove for IVC.
- The right anatomical lobe is much larger than the left.

Functional / physiological lobes

- Based on intrahepatic distribution of bile ducts, hepatic artery, and portal vein.
- Right and left true lobes are separated by Cantlie line, running approximately from the cystic notch/gallbladder fossa to the IVC.
- The functional right and left lobes are approximately equal in size.
- Each true lobe has its own primary branch of hepatic artery, portal vein, and hepatic duct.

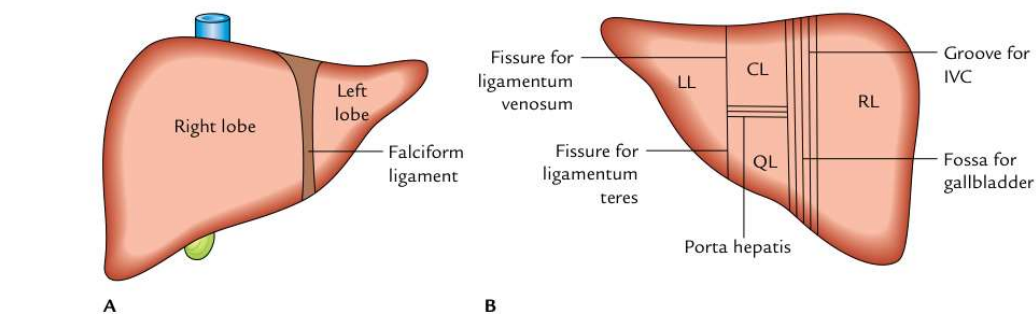


Fig. 8.6 Anatomical lobes of the liver: A, on the diaphragmatic surface of the liver; B, on the visceral surface of the liver (RL = right lobe, LL = left lobe, CL = caudate lobe, QL = quadrate lobe).

Source diagram: anatomical lobes on diaphragmatic and visceral surfaces.

Anatomical lobes are easy to see on the surface, but they do not accurately represent internal vascular and ductal territories. This is why surgical anatomy emphasizes functional lobes and segments.

Do not confuse these two divisions

Anatomical lobes are named by surface landmarks, while functional lobes are defined by the branching of the portal triad. In viva, first state which classification is being asked. Falciform ligament separates anatomical lobes, but Cantlie line separates functional lobes.

Anatomical lobe landmarks

- Right lobe: to the right of gallbladder fossa and fissures.
- Left lobe: to the left of fissures for ligamentum teres and venosum.
- Quadrate lobe: between gallbladder fossa and ligamentum teres below porta hepatis.
- Caudate lobe: between IVC groove and ligamentum venosum above porta hepatis.

Functional lobe landmarks

- Cantlie line passes from gallbladder fossa/cystic notch toward the IVC.
- It separates the liver according to blood supply and biliary drainage.
- This plane is used clinically in imaging and operative planning.

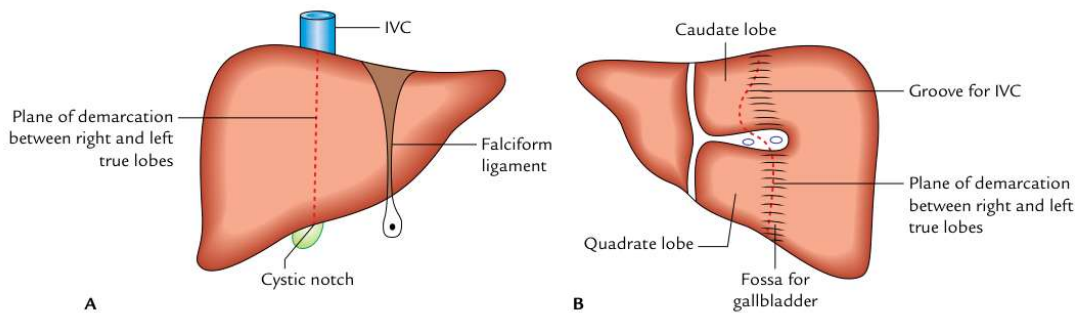


Fig. 8.7 True/physiological lobes of the liver: **A**, plane of demarcation on anterosuperior surface; **B**, visceral surface is shown by the interrupted redline.

Source diagram: true/physiological lobes and the plane of demarcation.

The red interrupted line represents the functional division. It passes through the gallbladder fossa anteriorly and the groove of the IVC posteriorly, making it more relevant in imaging and surgery than the falciform ligament.

4. Hepatic Segments

Concept of liver segments

Hepatic segments are surgically useful structural units of the liver. In Couinaud classification, there are eight segments numbered I to VIII. Segment I corresponds to the caudate lobe. Segments II-IV belong to the left hemiliver and segments V-VIII belong to the right hemiliver. Each segment has its own portal pedicle input and biliary drainage, while hepatic veins run intersegmentally.

Segment Map for Quick Recall

Segment group	Main idea
I	Caudate lobe; often considered special because it may receive supply/drainage from both sides.
II and III	Left lateral sector: superior and inferior parts.
IV	Left medial sector; often related to quadrate region.
V and VIII	Right anterior sector: inferior and superior parts.
VI and VII	Right posterior sector: inferior and superior parts.

Role of hepatic veins in segmental anatomy

The hepatic veins run between segments rather than inside individual segments. This is why they are called intersegmental veins. By contrast, the portal triad branches run into segments. Remember: portal triad defines the segment; hepatic veins separate the segments.

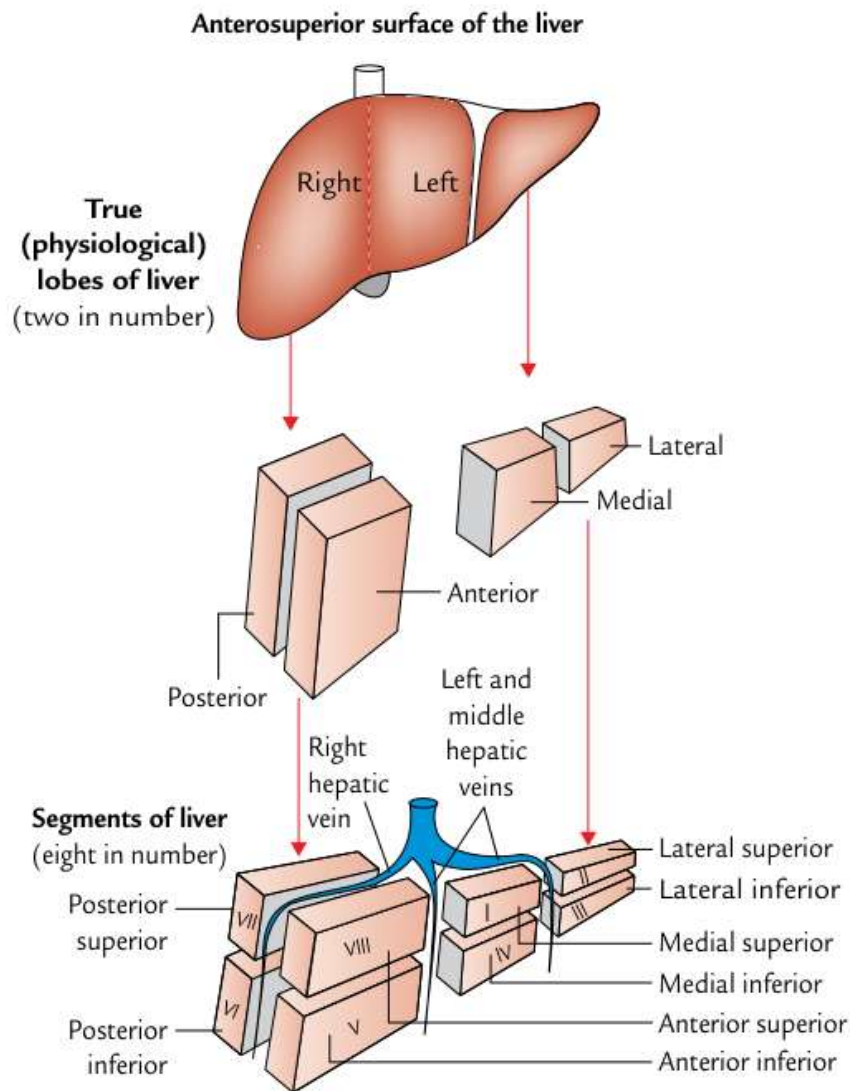


Fig. 8.8 Segments of the liver.

Source diagram: Couinaud segments of the liver.

This diagram shows the liver divided into eight segments. The right lobe is subdivided into anterior and posterior sectors, and the left lobe into medial and lateral sectors. This pattern explains why surgeons can remove diseased parts of liver while preserving the remaining vascular and biliary units.

Exam point: segmental resection

Because hepatic segments are functional units, segmental or sectoral resections can be planned around portal triad branches and hepatic veins. The liver also has a remarkable regenerative capacity; a large volume can be removed if the remaining liver is healthy.

5. Peritoneal Relations and Ligaments

Peritoneal coverage

Most of the liver is covered by peritoneum. Areas not covered by peritoneum include the bare area,

fossa for gallbladder, groove for IVC, fissure for ligamentum venosum, and porta hepatis. The bare area is clinically important because it is in direct contact with the diaphragm.

False ligaments (peritoneal folds)

- Falciform ligament: connects liver to anterior abdominal wall and diaphragm; contains ligamentum teres in its free margin.
- Coronary ligament: reflections around bare area; its layers continue laterally as triangular ligaments.
- Right triangular ligament: small peritoneal fold connecting right lobe to diaphragm.
- Left triangular ligament: fold connecting upper surface of left lobe to diaphragm.
- Lesser omentum: connects liver to lesser curvature of stomach and first part of duodenum.

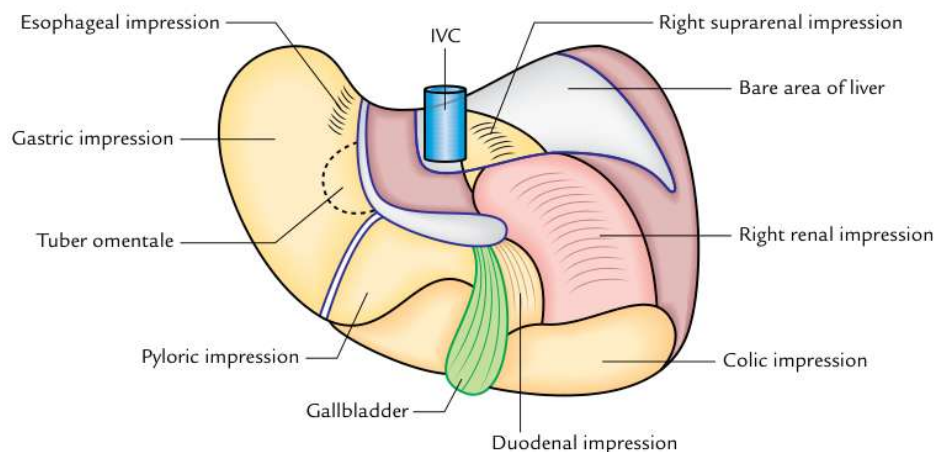
True ligaments (fetal remnants)

- Ligamentum teres hepatis: remnant of obliterated left umbilical vein; runs in fissure for ligamentum teres.
- Ligamentum venosum: remnant of ductus venosus; lies in fissure for ligamentum venosum.
- These true ligaments are not peritoneal folds in origin; they are fibrous remnants of fetal circulation.

6. Porta Hepatis and Visceral Relations

Porta hepatis

Porta hepatis is the gateway of the liver. It is a horizontal fissure on the visceral surface between the caudate and quadrate lobes. Structures entering include the portal vein, hepatic artery, lymphatics, and nerves. Structures leaving include the right and left hepatic ducts. The order from posterior to anterior is VAD: vein, artery, duct.



Visceral relations of the liver (relations of the inferior/visceral surface of the liver).

Source diagram: visceral relations of the liver.

The visceral surface bears impressions formed by neighbouring organs. The left lobe is related to stomach and esophagus; the right lobe is related to gallbladder, duodenum, right colic flexure, right kidney, and right suprarenal gland. These impressions help in orientation during gross anatomy and surgery.

Main Relations of the Liver

Area	Relations
Superior diaphragmatic surface	Diaphragm; through diaphragm to lungs and pleura on either side; heart and pericardium centrally.
Anterior surface	Xiphoid process and anterior abdominal wall.

Right lateral surface	Diaphragm; upper third related to lung and pleura; middle third to costodiaphragmatic recess; lower third to ribs.
Posterior surface	Abdominal esophagus, right suprarenal gland, IVC, and bare area.
Visceral surface	Stomach, duodenum, gallbladder, right colic flexure, right kidney, and right suprarenal gland.

7. Blood Supply, Venous Drainage, Lymphatics, and Nerves

Blood supply

- Portal vein supplies nutrient-rich venous blood from the gastrointestinal tract; it contributes the larger share of hepatic blood flow.
- Hepatic artery supplies oxygenated arterial blood and contributes a smaller but oxygen-rich component.
- Within the liver, branches of portal vein, hepatic artery, and bile duct travel together as portal triads.

Venous drainage

- Venous blood leaves the liver mainly through right, middle, and left hepatic veins.
- Hepatic veins drain directly into the inferior vena cava just below the diaphragm.
- The hepatic veins are intersegmental and are useful landmarks in imaging and surgery.

Lymphatic drainage

Superficial lymphatics from the anterior liver drain to hepatic nodes and then to coeliac nodes. Some posterior superficial lymphatics pass through the diaphragm to posterior mediastinal nodes. Deep lymphatics accompany the portal triads; some ascend through the caval opening, while others drain to nodes at the porta hepatis.

Nerve supply

The liver receives sympathetic fibers from the coeliac plexus and parasympathetic fibers from the hepatic branch of the anterior vagal trunk. Pain due to stretching of the capsule or inflammation of hepatic peritoneum travels mainly with sympathetic fibers and may be referred to the epigastrium or shoulder region.

abdominal walls.

with the capsule of the organ.

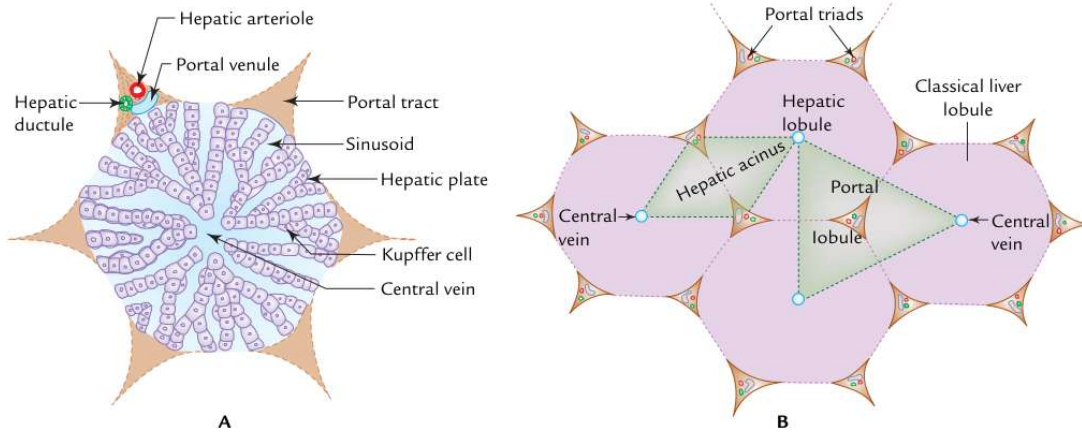


Fig. 8.11 A, classical liver lobule; B, functional units of liver, viz. portal lobule and hepatic acinus. (Source: A. Fig. 12.14, Page 239, *Textbook of Histology and a Practical Guide*, JP Gunasegaran. Copyright Elsevier 2010, All rights reserved. B. Fig. 12.17, Page 241, *Textbook of Histology and a Practical Guide*, JP Gunasegaran. Copyright Elsevier 2010, All rights reserved.)

Source diagram: microscopic structure of liver lobule and functional units.

The classical liver lobule has a central vein and portal triads at its corners. Blood flows from portal triads through sinusoids toward the central vein, while bile flows in the opposite direction through canaliculi toward bile ductules. This opposite flow is a high-yield concept.

8. Microscopic Structure - High-Yield Summary

Classical liver lobule

- Hexagonal unit centered on a central vein.
- Hepatocyte plates radiate toward the central vein.
- Portal triads lie at the periphery and contain a branch of portal vein, hepatic arteriole, and bile ductule.
- Sinusoids carry blood from periphery to central vein.

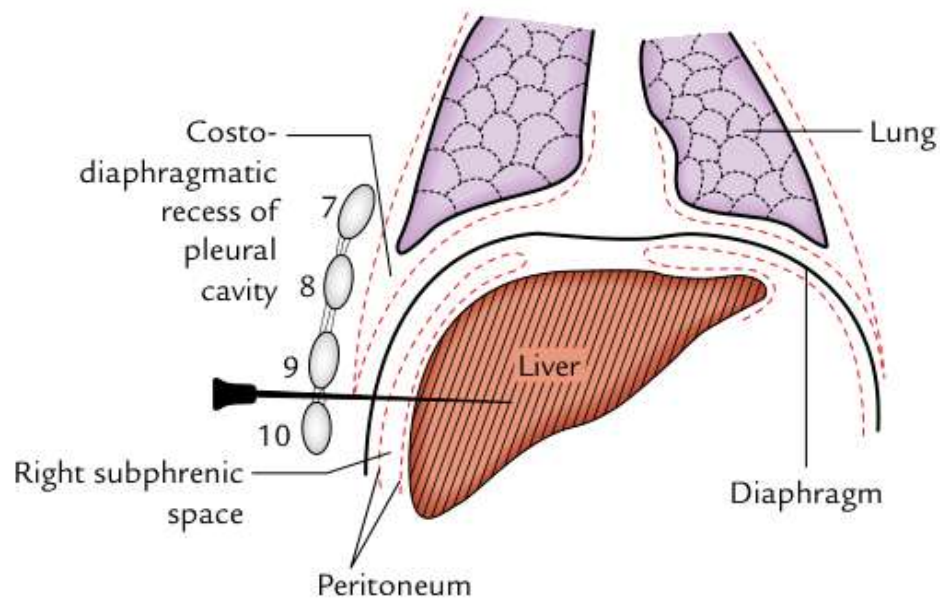
Portal lobule and hepatic acinus

- Portal lobule emphasizes bile drainage toward a bile ductule.
- Hepatic acinus emphasizes blood supply, oxygen gradient, and functional zonation.
- Kupffer cells are macrophages lining sinusoids and participate in phagocytosis.
- Space of Disse lies between sinusoidal endothelium and hepatocytes.

Direction of flow - high-yield line

Blood and bile flow in opposite directions. Blood from portal venules and hepatic arterioles enters sinusoids and moves toward the central vein. Bile produced by hepatocytes enters bile canaliculi and moves toward bile ductules in the portal triad. This opposite flow explains the arrangement of hepatocytes, sinusoids, and canaliculi.

9. Applied Anatomy



Source diagram: needle biopsy of liver.

The biopsy needle is inserted through an intercostal space while avoiding the pleura and lung. Clinically, the right midaxillary line at the 9th or 10th intercostal space is used, because a needle placed too high risks injuring the lung.

Clinical correlations

- Hepatomegaly: enlargement can make the liver palpable below the right costal margin; the inferior border descends during deep inspiration.
- Bare area of liver: because it contacts the diaphragm directly, infection from the liver can spread to the right pleural cavity through the diaphragm.
- Subphrenic abscess: may occur in relation to the bare area and can cause referred shoulder pain due to diaphragmatic irritation.
- Portal hypertension: vascular resistance in cirrhosis can cause portosystemic anastomoses to dilate, producing varices and ascites.
- Liver trauma: because of its size, vascularity, and protected but exposed subcostal position, it can bleed severely after abdominal trauma.
- Segmental liver surgery: knowledge of functional segments allows selective resection while preserving the remaining liver tissue.

10. Quick Revision Table

Golden Points

Question	Answer
Largest gland in the body	Liver
Main functional division	Right and left physiological lobes divided by Cantlie line
Gateway of liver	Porta hepatis
Order at porta hepatis from posterior to anterior	Vein, artery, duct (VAD)
Non-peritoneal area on posterior surface	Bare area of liver

Major venous drainage	Right, middle, and left hepatic veins into IVC
Functional units for surgery	Hepatic segments (Couinaud segments I-VIII)
Remnant of left umbilical vein	Ligamentum teres hepatis
Remnant of ductus venosus	Ligamentum venosum
Cell responsible for phagocytosis in sinusoids	Kupffer cell

Last-minute viva cues

Always distinguish anatomical lobes from functional lobes. Anatomical lobes are surface-based, while functional lobes follow vessels and ducts. For porta hepatis, remember VAD from posterior to anterior. For microscopic anatomy, remember: blood flows from portal triad to central vein, while bile flows from hepatocytes toward bile ductules in the portal triad.

How to write a long answer

- Start with definition, location, and functions.
- Describe surfaces, borders, and lobes with diagrams.
- Add peritoneal ligaments and porta hepatis.
- Finish with blood supply, venous drainage, lymphatics, nerves, and applied anatomy.

Common mistakes to avoid

- Do not call anatomical lobes the same as functional lobes.
- Do not forget that hepatic veins drain into the IVC, not the portal vein.
- Do not reverse the direction of bile and blood flow in the lobule.
- Do not miss the clinical importance of bare area and liver biopsy.

Diagram checklist for exams

Draw a simple anterior liver showing right and left lobes, gallbladder, falciform ligament, and IVC. For visceral surface, show the H-shaped arrangement: fissure for ligamentum teres, fissure for ligamentum venosum, gallbladder fossa, IVC groove, and porta hepatis. Label caudate and quadrate lobes clearly.

Source base: Vishram Singh, Textbook of Anatomy: Abdomen and Lower Limb, Volume II. Diagrams are adapted from the uploaded source PDF for study use.